

Model Lab

Every prior, same slate — 2026-09-27

SERVED PRIOR

Served (full season)

PRIORS COMPARED

11

HELD-OUT PA

37,547

Held-Out Scores

Fitted on plate appearances before 2026-08-20, scored on the 37,547 that came after (actual HR rate 2.98%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5821	1.57×	0.02884	0.13311	0.00497
KNN comps Prior is the pooled rate of the nearest comparables.	0.5904	1.44×	0.02883	0.13265	0.00541
SVR Support-vector regression on the same feature matrix.	0.5858	1.47×	0.02891	0.13389	0.00815
Random forest Forest regression; a learned similarity metric.	0.5864	1.49×	0.02888	0.13315	0.00688
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5904	1.44×	0.02883	0.13265	0.00541
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5904	1.44×	0.02883	0.13265	0.00541
Linear regression Ordinary least squares on the same features.	0.5885	1.51×	0.02886	0.13311	0.00552
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5901	1.55×	0.02891	0.13325	0.00760
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5887	1.53×	0.02886	0.13295	0.00662
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5870	1.53×	0.02891	0.13346	0.00768

Blend weights (forest variant): KNN 1.00, SVR 0.00, RF 0.00

Shrinkage K per prior: league 257, knn 257, svr 257, rf 257, core 257, ensemble 257, linear 257, logistic 257, nn 257, xgb 257 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (4,288 PA, actual HR rate 2.50%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5301	1.49×	0.02435	0.11710	0.01018
KNN comps Prior is the pooled rate of the nearest comparables.	0.5391	1.12×	0.02452	0.11981	0.01599
SVR Support-vector regression on the same feature matrix.	0.5456	1.03×	0.02508	0.12771	0.02860
Random forest Forest regression; a learned similarity metric.	0.5524	1.21×	0.02486	0.12290	0.02418
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5391	1.12×	0.02452	0.11981	0.01599
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5391	1.12×	0.02452	0.11981	0.01599
Linear regression Ordinary least squares on the same features.	0.5495	1.12×	0.02477	0.12361	0.02120
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5490	1.12×	0.02506	0.12346	0.02690
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5570	1.21×	0.02466	0.12080	0.02011
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5585	1.03×	0.02510	0.12473	0.02972

Slate Agreement

Served (full season) · 6/6 shared 1/5 hit

Willson Contreras
Pete Alonso
Yordan Alvarez
Rafael Flores Jr.
Munetaka Murakami
Griffin Conine ✓

KNN comps · 6/6 shared 1/5 hit

Willson Contreras
Pete Alonso
Yordan Alvarez
Rafael Flores Jr.
Munetaka Murakami
Griffin Conine ✓

Random forest · 6/6 shared 1/5 hit

Pete Alonso
Yordan Alvarez
Willson Contreras
Rafael Flores Jr.
Munetaka Murakami
Griffin Conine ✓

Forest blend · 6/6 shared 1/5 hit

Willson Contreras
Pete Alonso
Yordan Alvarez
Rafael Flores Jr.
Munetaka Murakami
Griffin Conine ✓

Logistic regression · 6/6 shared 1/5 hit

Flat league · 5/6 shared 1/5 hit

20.6% Pete Alonso 19.3%
20.5% Willson Contreras 19.0%
20.4% Yordan Alvarez 18.9%
19.2% Jeremy Peña 18.5%
19.1% Rafael Flores Jr. 17.4%
18.7% Griffin Conine ✓ 17.3%

SVR · 6/6 shared 1/5 hit

20.7% Yordan Alvarez 21.0%
20.5% Pete Alonso 21.0%
20.3% Willson Contreras 20.6%
19.3% Rafael Flores Jr. 20.4%
18.9% Munetaka Murakami 19.6%
18.8% Griffin Conine ✓ 19.3%

Core (KNN+SVR) · 6/6 shared 1/5 hit

20.8% Willson Contreras 20.7%
20.7% Pete Alonso 20.5%
20.7% Yordan Alvarez 20.3%
19.6% Rafael Flores Jr. 19.3%
19.2% Munetaka Murakami 18.9%
19.1% Griffin Conine ✓ 18.8%

Linear regression · 6/6 shared 1/5 hit

20.7% Willson Contreras 20.9%
20.5% Pete Alonso 20.5%
20.3% Yordan Alvarez 20.5%
19.3% Rafael Flores Jr. 19.7%
18.9% Griffin Conine ✓ 19.0%
18.8% Munetaka Murakami 18.8%

Willson Contreras	21.6%	Neural net (torch) · 6/6 shared 1/5 hit	
Yordan Alvarez	21.2%	Willson Contreras	21.0%
Pete Alonso	20.9%	Yordan Alvarez	20.8%
Rafael Flores Jr.	19.7%	Pete Alonso	20.6%
Griffin Conine ✓	19.3%	Rafael Flores Jr.	19.8%
Munetaka Murakami	19.0%	Griffin Conine ✓	19.3%
		Munetaka Murakami	19.1%
XGBoost · 6/6 shared 1/5 hit			
Pete Alonso	20.8%		
Willson Contreras	20.8%		
Yordan Alvarez	20.7%		
Rafael Flores Jr.	19.7%		
Munetaka Murakami	19.6%		
Griffin Conine ✓	19.2%		

Hitter by Hitter

Hitter	Served (full season)	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression
19. Shohei Ohtani Los Angeles Dodgers · vs LHP · 30 HR / 615 PA	16.7%	15.2%	16.5%	17.1%	17.0%	16.5%	16.5%	16.8%	17.1%
20. Matt Olson Atlanta Braves · vs RHP · 41 HR / 696 PA	16.6%	15.4%	16.5%	17.3%	16.9%	16.5%	16.5%	16.6%	17.3%
21. Francisco Lindor New York Mets · vs LHP · 20 HR / 464 PA	16.5%	16.0%	16.5%	16.6%	16.4%	16.5%	16.5%	16.3%	16.4%
22. Andrés Chaparro Washington Nationals · vs LHP · 11 HR / 224 PA	16.4%	15.4%	16.3%	16.3%	16.2%	16.3%	16.3%	15.9%	15.8%
23. Elly De La Cruz Cincinnati Reds · vs LHP · 30 HR / 632 PA	16.4%	15.6%	16.3%	16.8%	16.7%	16.3%	16.3%	16.4%	16.7%
24. Hunter Goodman Colorado Rockies · vs LHP · 41 HR / 570 PA	16.3%	15.6%	16.3%	16.9%	16.6%	16.3%	16.3%	16.5%	16.8%
25. Riley Greene Detroit Tigers · vs RHP · 21 HR / 588 PA	16.2%	14.9%	15.9%	16.2%	16.1%	15.9%	15.9%	15.9%	16.3%
26. Jo Adell slate Cleveland Guardians · vs LHP · 22 HR / 657 PA	16.0%	15.4%	16.1%	16.5%	15.7%	16.1%	16.1%	16.0%	16.4%
27. Lawrence Butler Athletics · vs RHP · 17 HR / 536 PA	15.9%	15.9%	16.0%	15.8%	15.7%	16.0%	16.0%	16.0%	15.7%
28. Jake Burger Texas Rangers · vs LHP · 28 HR / 616 PA	15.9%	14.8%	15.8%	16.1%	15.8%	15.8%	15.8%	15.8%	16.1%
29. Junior Caminero Tampa Bay Rays · vs RHP · 42 HR / 700 PA	15.9%	15.5%	16.0%	17.4%	16.5%	16.0%	16.0%	16.2%	16.5%
30. Dylan Beavers Baltimore Orioles · vs RHP · 9 HR / 363 PA	15.8%	15.2%	15.8%	15.5%	15.5%	15.8%	15.8%	15.6%	15.4%
31. Luis García Jr. New York Yankees · vs RHP · 30 HR / 543 PA	15.5%	14.9%	15.5%	16.2%	16.1%	15.5%	15.5%	15.8%	16.1%
32. CJ Abrams Washington Nationals · vs LHP · 33 HR / 650 PA	15.5%	14.6%	15.6%	15.7%	15.8%	15.6%	15.6%	15.4%	15.5%
33. Heriberto Hernández	15.5%	14.2%	15.5%	15.5%	15.6%	15.5%	15.5%	15.4%	15.5%

Hitter	Served (full season)	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression
Miami Marlins · vs RHP · 24 HR / 501 PA									
34. Zach Neto Los Angeles Angels · vs LHP · 27 HR / 699 PA	15.4%	14.2%	15.5%	15.5%	15.4%	15.5%	15.5%	15.0%	15.4%
35. Adley Rutschman Boston Red Sox · vs RHP · 12 HR / 407 PA	15.2%	14.9%	15.1%	14.8%	14.5%	15.1%	15.1%	14.9%	14.6%
36. Eduardo Valencia Detroit Tigers · vs RHP · 7 HR / 134 PA	15.2%	14.1%	15.4%	15.1%	15.3%	15.4%	15.4%	15.1%	15.3%
37. Angel Martínez Cleveland Guardians · vs LHP · 18 HR / 412 PA	15.1%	15.6%	15.1%	15.6%	15.4%	15.1%	15.1%	15.3%	15.5%
38. Spencer Jones New York Yankees · vs RHP · 11 HR / 289 PA	14.9%	13.7%	15.0%	13.9%	14.1%	15.0%	15.0%	14.0%	14.1%
39. Shea Langeliers Athletics · vs RHP · 23 HR / 464 PA	14.9%	13.9%	15.0%	16.0%	15.5%	15.0%	15.0%	15.2%	15.8%
40. Tommy Pham Chicago White Sox · vs LHP · 1 HR / 31 PA	14.8%	14.7%	15.2%	9.8%	14.1%	15.2%	15.2%	14.9%	13.3%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.